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Chapter 1: Database Management System DBMS (MCQ)

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1. Which of the following SQL statement is used to DELETE rows from a database table?
a. DELETE
b. REMOVE
c. DROP
d. CLEAR

ans: a) DELETE

2. A field that is used to uniquely define a particular record in a table is called:
a. Primary Key
b. Entity
c. Relationship
d. Constraints

ans: a) Primary Key

3. ... is not the data type of MS-Access.
a. Number
b. Text
c. Memo
d. Form

ans: d. Form

4. ... is the DBMS.
a. Fox-Pro
b. MS Excel
c. Lotus-123
d. All of the above

ans: a. Fox-Pro

5. The primary key does not accept . . .
a. Text
b. Number
c. Null Value
d. None of above

ans: c. Null Value

6. In MS Access, we enter data using . . .
a. Form
b. Table
c. Report
d. Both a and b

ans: d. Both a and b

7. Date/Time occupies ... bytes of memory.
a. 4
b. 2
c. 8
d. 16

ans: c. 8

8. A relational database consists of a collection of
a. Tables
b. Fields
c. Records
d. Keys

ans: a. Tables

9. The term _____ is used to refer to a row.
a. Attribute
b. Tuple
c. Field
d. Instance

ans: b. Tuple

10. The term attribute refers to a _____ of a table.
a. Record
b. Column
c. Tuple
d. Key

ans: b. Column

11. CREATE TABLE employee (name VARCHAR, id INTEGER) – What type of statement is this?
a. DML
b. DDL
c. View
d. Integrity constraint

ans: b. DDL

12. In the given query which keyword has to be inserted? INSERT INTO employee (4, Alok, 2000);
a. Table
b. Values
c. Relation
d. Field

ans: b. Values

13. In the _____ normal form, a composite attribute is converted to individual attributes.
a. First
b. Second
c. Third
d. Fourth

ans: a. First

14. With SQL, how do you select all the records from a table named "Persons" where the value of the column "FirstName" ends with an "a"?
a. SELECT * FROM Persons WHERE FirstName='a'
b. SELECT * FROM Persons WHERE FirstName LIKE 'a%'
c. SELECT * FROM Persons WHERE FirstName LIKE '%a'
d. SELECT * FROM Persons WHERE FirstName='%a%'

ans: c. SELECT * FROM Persons WHERE FirstName LIKE '%a'

15. The UPDATE SQL clause can
a. update only one row at a time
b. update more than one row at a time
c. delete more than one row at a time
d. delete only one row at a time

ans: b. update more than one row at a time

16. Which SQL keyword is used to retrieve a maximum value?
a. MOST
b. TOP
c. MAX
d. UPPER

ans: c. MAX

17. Which of the following is not a DDL command?
a. UPDATE
b. TRUNCATE
c. ALTER
d. None of the Mentioned

ans: a. UPDATE

18. It is better to use DBMS than a file when there are
a. Multiple users wish to access the data
b. Stringent real-time requirements
c. Complex relationships among data
d. All of the above

ans: d. All of the above

19. DBMS is type of ..... software.
a. Application software
b. System software
c. General software
d. Both b and c

ans: b. System software

20. To which of the following the term DBA referred?
a. Data bank administrator
b. Data administrator
c. Database administrator
d. All of the above

ans: c. Database administrator

21. Duplication of data in database is known as
a. Data Integrity
b. Data redundancy
c. Data inconsistency
d. None of the above

ans: b. Data redundancy

22. Which of the following enables us to view data from a table based on a specific criterion?
a. Form
b. Macro
c. Query
d. Report

ans: c. Query

23. Which of the following is used to establish an association between related tables?
a. Line
b. Primary Key
c. Relationship
d. Records

ans: c. Relationship

24. Which key is used to uniquely identify each record in database?
a. Unique Key
b. Primary Key
c. Key Record
d. Field Name

ans: b. Primary Key

25. Which form simplifies and ensures that there are minimal data aggregates and repetitive groups:
a. 1NF
b. 2NF
c. 3NF
d. All of the mentioned

ans: a. 1NF

26. A database language concerned with the definition of the whole database structure and schema is:
a. DCL (Data Control Language)
b. DML (Data Manipulation Language)
c. DDL (Data Definition Language)
d. All of the above

ans: c. DDL (Data Definition Language)

27. Which of the following is not a database model?
a. Object Oriented Database Model
b. Network Database Model
c. Relational Database Model
d. Waterfall Database Model

ans: d. Waterfall Database Model

28. The overall description of a database is called
a. Data integrity
b. Database Schema
c. Data Definition
d. Data Manipulation

ans: b. Database Schema

29. Which of the following is not a DDL statement?
a. CREATE
b. SELECT
c. ALTER
d. DROP

ans: b. SELECT

30. Which of the following is not a DML statement?
a. CREATE
b. SELECT
c. INSERT
d. DELETE

ans: a. CREATE

31. In a relational model, relations are termed as
a. Tuples
b. Attributes
c. Tables
d. Rows

ans: c. Tables

32. In Hierarchical Model records are organized as
a. Table
b. List
c. Links
d. Tree

ans: d. Tree

33. What is used for database security?
a. Data encryption
b. Password
c. Finger print
d. All of the above

ans: a. Data encryption

34. Prevention of access to the database by unauthorized.
a. Integrity
b. Security
c. Productivity
d. Reliability

ans: b. Security

35. SQL stands for
a. Simple Query Language
b. Structured Quarter Language
c. Structure Question Language
d. Structured Query Language

ans: d. Structured Query Language

36. No. of Tuples in Relation is called
a. Degree
b. Attribute
c. Cardinality
d. Vertex

ans: c. Cardinality

37. What is the advantage of distributed database over centralized database?
a. Software cost
b. Software complexity
c. Slow response
d. Modular growth

ans: d. Modular growth

38. Which one of the following refers to the copies of the same data (or information) occupy the memory space at multiple places.
a. Data Repository
b. Data Inconsistency
c. Data Mining
d. Data Redundancy

ans: d. Data Redundancy

39. What does "MAC" stand for in MAC Address?
a. Mandatory Access Control
b. Media Access Control
c. Micro Access Control
d. Media Access Certificate

ans: b. Media Access Control

40. _____ refers to the sending, receiving, and processing of information by electronic means.
a. Computer

b. Communication
c. Software
d. Program

ans: b. Communication

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CHAPTER 1: DATABASE MANAGEMENT SYSTEM (DBMS)

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SHORT ANSWER QUESTIONS (5 Marks)

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1. What is normalization? Explain 2NF and/or 3NF with examples.
Asked: 9 times
(2082 Q.No.10OR | 2081 GIE Set A Q.No.10 | 2081 Q.No.10OR | 2080 GIE Set B Q.No.10 | 2079 Set A Q.No.10OR | 2077 Set D Q.No.4 | 2076 GIE Set A Q.No.7 | 2076 GIE Set B Q.No.7 | 2074 Set A Q.No.8)

2. Explain the Relational Database Model with example.
Asked: 4 times
(2080 GIE Set A Q.No.10OR | 2080 Q.No.10 | 2079 GIE Set A Q.No.10 | 2073 Set D Q.No.9)

3. Differentiate between centralized and distributed database system.
Asked: 4 times
(2081 GIE Set B Q.No.10OR | 2079 Set A Q.No.10 | 2076 GIE Set B Q.No.10 | 2075 Set A Q.No.8)

4. What are the roles/functions of a Database Administrator (DBA)? Explain.
Asked: 3 times
(2079 GIE Set B Q.No.10 | 2078 Set C Q.No.10 | 2076 GIE Set B Q.No.8)

5. Write any three differences between DDL and DML. Give examples of each.
Asked: 2 times
(2082 Q.No.10 | 2076 Q.No.8)

6. Describe DDL statements with examples in SQL.
Asked: 2 times
(2079 GIE Set A Q.No.10OR | 2079 GIE Set B Q.No.10OR)

7. Write SQL DDL statements to create a table with given schema/attributes.
Asked: 2 times
(2081 GIE Set A Q.No.10OR | 2080 Q.No.10OR)

8. List the different types of database models. Explain Relational Database model in brief.
Asked: 2 times
(2081 GIE Set B Q.No.10 | 2078 Set C Q.No.8)

9. Evaluate the advantages/importance of DBMS compared to traditional file-based systems.
Asked: 2 times

(2081 Q.No.10 | 2080 GIE Set A Q.No.10)

10. Explain E-R diagram / entity, attribute and relationship in DBMS.
Asked: 2 times
(2078 Set C Q.No.9 | 2076 GIE Set A Q.No.8)

11. What is the importance of database security in DBMS? Describe.
Asked: 1 time
(2080 GIE Set B Q.No.10OR)

12. Explain the different types of data integrity in DBMS.
Asked: 1 time
(2076 Q.No.7)

LONG ANSWER QUESTIONS (8 Marks)

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1. Define normalization. Explain 2NF and 3NF with examples. / What is normalization? Write its advantages and explain 2NF and 3NF with examples.
Asked: 3 times
(2078 Set C Q.No.4 | 2075 Q.No.5 | 2075 Set B Q.No.5)

2. What do you understand by data integrity? Why is it important while designing a database? State and describe different types of data integrity.
Asked: 1 time
(2062 Q.No.2OR)

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Chapter 2: Data Communication & Networking (MCQ)

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41. Which refers to the communication media?
a. UTP cable
b. Fiber optic cable
c. Satellite
d. All of the above

ans: d. All of the above

42. Which one is not a network topology?

a. Star
b. Ring
c. Client Server
d. BUS

ans: c. Client Server

43. Which is not a guided media?
a. UTP Cable
b. STP Cable
c. Satellite
d. Fiber Optic

ans: c. Satellite

44. Which device is necessary to operate the Internet?
a. CD-ROM
b. ISP
c. Modem
d. All of the above

ans: b. ISP

45. In a digital data transmission of computer networking, what is Mbps?
a. Mega Bits Per Second
b. Mega Bytes Per Second
c. Mega Bands Per Second
d. Media Bits Per Second

ans: a. Mega Bits Per Second

46. In client server network architecture, which statement is not true?
a. Secured and reliable network architecture
b. Data traffic is very high because of broad architecture
c. It is less expensive architecture
d. Network controlling and management is easier

ans: b. Data traffic is very high because of broad architecture

47. Which topology is more reliable and secured for a LAN?
a. Star
b. Bus
c. Ring
d. Mesh

ans: d. Mesh

48. What is the correct order of OSI reference model of a computer network from bottom to top?
a. Physical, datalink, network, transport, session, application and presentation
b. Physical, datalink, network, transport, presentation, session, and application
c. Physical, datalink, transport, network, session, presentation and application
d. Physical, datalink, network, transport, session, presentation and application

ans: d. Physical, datalink, network, transport, session, presentation and application

49. Radio broadcast system is an example of transmission.
a. Simplex
b. Half-Duplex

c. Full-Duplex
d. None of the above

ans: a. Simplex

50. Communication between computer and monitor is Transmission.
a. Simplex
b. Half-Duplex
c. Full-Duplex
d. None of the above

ans: a. Simplex

51. Walkie-Talkie communication is an example of transmission.
a. Simplex
b. Half-Duplex
c. Full-Duplex
d. None of the above

ans: b. Half-Duplex

52. Transmission mode in which data can transmit in both directions at the same time.
a. Simplex
b. Half-Duplex
c. Full-Duplex
d. None of the above

ans: c. Full-Duplex

53. Transmission mode in which capacity of transmission medium is shared all the time is
a. Simplex
b. Half-Duplex
c. Full-Duplex
d. None of the above

ans: b. Half-Duplex

54. Computer network within building, campus or nearby building is
a. LAN
b. MAN
c. WAN
d. PAN

ans: a. LAN

55. Computer network that span countries, continent or the whole world is
a. LAN
b. MAN
c. WAN
d. PAN

ans: c. WAN

56. Transmission medium that provides the physical conduit from one device to another is
a. Guided

b. Unguided
c. Both A and B
d. None of the above

ans: a. Guided

57. Wireless medium is also called _____ medium.
a. Guided
b. Unguided
c. Unbounded
d. Both B and C

ans: b. Unguided

58. Transmission medium in which signal transmit in the form of light is
a. Twisted pair cable
b. Communication satellite
c. Coaxial cable
d. Fiber-optic cable

ans: d. Fiber-optic cable

59. _____ cable can transmit data in higher rate than _____ cable.
a. Twisted pair ; coaxial
b. Coaxial ; Fiber-optic
c. Coaxial ; Twisted pair
d. Twisted pair ; Fiber-optic

ans: b. Coaxial ; Fiber-optic

60. _____ media transmit the signal without using physical conductor.
a. Bounded
b. Unbounded
c. Unguided
d. Both B and C

ans: b. Unbounded

61. The OSI model consists of _____ layers.
a. 3
b. 5
c. 7
d. 9

ans: c. 7

62. The _____ layer is closest to transmission medium.
a. Physical
b. Data link
c. Transport
d. Network

ans: a. Physical

63. The end-to-end delivery of the entire message is the responsibility of _____ layer.
a. Data link
b. Network
c. Transport
d. Presentation

ans: c. Transport

64. The _____ layer enables the user to access network.
a. Presentation
b. Session
c. Network
d. Application

ans: d. Application

65. In fiber optics, the signal source is
a. Radio waves
b. Infrared waves
c. Light waves
d. Very low frequency waves

ans: c. Light waves

66. Geosynchronous satellite complete one orbit in
a. 1 hour
b. 1 day
c. 1 month
d. 1 year

ans: b. 1 day

67. _____ is a type of transmission impairment in which outside source such as crosstalk corrupts the signal.
a. Attenuation
b. Noise
c. Distortion
d. Decibel

ans: b. Noise

68. _____ is a loss of signal's energy due to the resistance of the medium.
a. Attenuation
b. Noise
c. Distortion
d. Decibel

ans: a. Attenuation

69. _____ topology uses multipoint connection.
a. Star
b. Bus
c. Ring
d. Mesh

ans: b. Bus

70. _____ topology requires hub or central controller.
a. Star
b. Bus
c. Ring
d. Mesh

ans: a. Star

71. Mesh topology with 6 nodes requires
a. 6
b. 10
c. 12
d. 15

ans: d. 15

72. Topology that uses point-to-point connection
a. Star
b. Mesh
c. Ring
d. All of the above

ans: b. Mesh

73. Size of IP address in IPv4 is
a. 4 Bytes
b. 8 Bytes
c. 16 Bytes
d. None of the above

ans: a. 4 Bytes

74. Size of IP address in IPv6 is
a. 32 Bits
b. 64 Bits
c. 128 Bits
d. None of the above

ans: c. 128 Bits

75. Size of subnet mask is
a. 32 Bits
b. 64 Bits
c. 128 Bits
d. None of the above

ans: a. 32 Bits

76. Size of MAC address is
a. 32 Bits
b. 48 Bits
c. 64 Bits

d. 128 Bits

ans: b. 48 Bits

77. Private LAN/MAN of an organization that is accessible to organizational users only is
a. Internet
b. Intranet
c. Extranet
d. All of the above

ans: b. Intranet

78. Which one is a remote login tool.
a. LogMeIn
b. TeamViewer
c. SplashTop
d. All of the above

ans: d. All of the above

79. _____ regenerates an incoming signal before retransmitting it.
a. HUB
b. Modem
c. Repeater
d. Router

ans: c. Repeater

80. Wi-Fi uses _____ gigahertz radio bands.
a. 2.4
b. 5
c. Both A and B
d. None of the above

ans: c. Both A and B

81. _____ command verifies IP-level connectivity to another TCP/IP computer by sending ICMP echo request messages.
a. Ipconfig
b. Ping
c. Both A and B
d. None of the above

ans: b. Ping

82. _____ command is used to display the basic TCP/IP configuration for all adapters.
a. Ipconfig
b. Ping
c. Both A and B
d. Decibel

ans: a. Ipconfig

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CHAPTER 2: DATA COMMUNICATION AND NETWORKING

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SHORT ANSWER QUESTIONS (5 Marks)

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1. What is transmission media? Describe/explain the types of transmission media with examples.
   Asked: 6 times
   (2076 GIE Set B Q.No. 9 | 2075 Set A Q.No. 10 | 2075 Set B Q.No. 10 | 2074 Set A Q.No. 10 | 2074 Set B Q.No. 9 | 2071 Set C Q.No. 10)

2. Compare/differentiate star and ring topology with features/pros and cons.
   Asked: 2 times
   (2079 GIE Set B Q.No. 11 OR | 2077 Set D Q.No. 5)

3. Define network topology. Explain the Ring topology with diagram/features.
   Asked: 2 times
   (2076 Q.No. 9 | 2075 GIE Q.No. 9)

4. Differentiate between LAN and WAN.
   Asked: 2 times
   (2076 GIE Set A Q.No. 10 | 2074 Set B Q.No. 10)

5. Differentiate between peer-to-peer and client/server network.
   Asked: 2 times
   (2075 Set A Q.No. 9 | 2073 Supp Q.No. 9)

6. Describe simplex, half duplex, and full duplex with examples.
   Asked: 2 times
   (2072 Set C Q.No. 11 | 2070 Supp Q.No. 10)

7. What is Networking? List the advantages of Networking.
   Asked: 2 times
   (2072 Set D Q.No. 9 | 2070 Set C Q.No. 8)

8. Differentiate between bounded and unbounded transmission media.
   Asked: 1 time
   (2075 GIE Q.No. 10)

9. Describe the Bus and Star topology with suitable diagram.
   Asked: 1 time
   (2073 Set C Q.No. 9)

10. What is computer network? List out hardware equipments to establish the computer network.
    Asked: 1 time
    (2076 GIE Set A Q.No. 9)

11. Describe the wireless network system. List out devices and equipments necessary for Wi-Fi network.

Asked: 1 time

(2076 Q.No. 10)

12. Describe the class C IP address with an example.

Asked: 1 time

(2079 GIE Set B Q.No. 11)

13. Explain about coaxial cable and fiber optics cable.

Asked: 1 time

(2078 Set C Q.No. 11)

LONG ANSWER QUESTIONS (8 Marks)

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1. Explain client-server and peer-to-peer network architecture with diagram. Write advantages and disadvantages of each.

Asked: 2 times

(2081 GIE Set B Q.No. 15 | 2079 GIE Set A Q.No. 15)

2. Compare star and bus network topology. Which network cable is suitable for the school's/local area network? Justify.

Asked: 2 times

(2081 Q.No. 15 | 2080 GIE Set B Q.No. 15)

3. What is transmission medium? Explain its major types with advantages and disadvantages.

Asked: 1 time

(2082 Q.No. 15)

4. Why do network designers prefer switch in LAN? Why do people prefer star topology in organizations like a bank? Give reasons.

Asked: 1 time

(2081 GIE Set A Q.No. 15)

5. Describe CAT-6 and optical fiber cable features. Which is more suitable to develop the local area network?

Asked: 1 time

(2080 GIE Set A Q.No. 15)

6. What is network topology? Differentiate between guided (wire) and unguided (wireless) media with examples.

Asked: 1 time

(2080 Q.No. 15)

7. Compare star and ring topology with pros and cons. Which data communication cable is more appropriate to design a local area network? Describe.

Asked: 1 time

(2079 Set A Q.No. 15)

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Chapter 4: Programming Concepts (C, Functions, Pointers, File Handling) (MCQ)

==========

101. The _____ statement is used to delete a table.
a. DROP TABLE
b. DELETE TABLE
c. DEL TABLE
d. REMOVE TABLE

ans: a. DROP TABLE

102. Which program copies the databases from one server to another?
a. Mysqlcopydb
b. Mysqldbcopy
c. Mysqlflushdb
d. Mysqldbflush

ans: a. Mysqlcopydb

103. Which among the following have the maximum bytes?
a. Varchar
b. Char
c. Both Varchar and Char
d. Text Type

ans: d. Text Type

104. In C programming, the main() function type is
a. int
b. void
c. float
d. char

ans: a. int

105. Which of the following is not a C data type?
a. int
b. float
c. real
d. char

ans: c. real

106. Pointer is a variable whose value is
a. Integer
b. Address
c. Float
d. Character

ans: b. Address

107. Function returning no value is
a. void function
b. int function

c. main function
d. char function

ans: a. void function

108. Which operator is used to access value at the address stored in a pointer?
a. &
b. *
c. %
d. ->

ans: b. *

109. Which operator is used to find the address of variable?
a. &
b. *
c. %
d. ->

ans: a. &

110. fopen() is used to
a. Open file
b. Create file
c. Close file
d. None of the above

ans: a. Open file

111. fclose() is used to
a. Open file
b. Close file
c. Create file
d. None of the above

ans: b. Close file

112. feof() function is used to
a. Open file
b. Check end of file
c. Close file
d. None of the above

ans: b. Check end of file

113. Which of the following is not a loop in C?
a. for
b. do-while
c. while
d. until

ans: d. until

114. Which of the following is a logical operator in C?
a. &&
b. +
c. -

d. *

ans: a. &&

115. Which of the following is a conditional operator in C?
a. &&
b. ?:
c. ==
d. !=

ans: b. ?:

116. The size of integer data type in C is
a. 2 bytes
b. 4 bytes
c. 8 bytes
d. 1 byte

ans: b. 4 bytes

117. Array index in C starts with
a. 0
b. 1
c. -1
d. None of the above

ans: a. 0

118. Function with same name but different arguments is called
a. Overloading
b. Overriding
c. Recursion
d. None of the above

ans: a. Overloading

119. Recursion is a process in which function calls
a. Another function
b. Itself
c. Main function
d. Library function

ans: b. Itself

120. The output of printf("%d", sizeof(char)); is
a. 1
b. 2
c. 4
d. Depends on compiler

ans: a. 1

121. The output of printf("%d", sizeof(int)); is
a. 1

b. 2
c. 4
d. 8

ans: c. 4

122. The output of printf("%d", sizeof(float)); is
a. 2
b. 4
c. 8
d. Depends on compiler

ans: b. 4

123. Which of the following function can convert string to integer?
a. itoa
b. atoi
c. atol
d. atof

ans: b. atoi

124. Which function is used to copy one string to another?
a. strcpy
b. strcat
c. strcmp
d. strrev

ans: a. strcpy

125. Which function is used to compare two strings?
a. strcpy
b. strcat
c. strcmp
d. strrev

ans: c. strcmp

126. Which function is used to concatenate two strings?
a. strcpy
b. strcat
c. strcmp
d. strrev

ans: b. strcat

127. Which function returns the length of string?
a. strlen
b. strcat
c. strcpy
d. strcmp

ans: a. strlen

128. Which of the following is not a storage class in C?

a. auto
b. register
c. static
d. inline

ans: d. inline

129. Which operator is used to access members of structure through pointer?
a. .
b. ->
c. &
d. *

ans: b. ->

130. Which of the following is used to define a macro?
a. #define
b. #include
c. #macro
d. #header

ans: a. #define

131. Which of the following is not valid identifier in C?
a. _var1
b. var_2
c. 2var
d. var3

ans: c. 2var

132. Which of the following is correct about C language?
a. High level language
b. Assembly language
c. Low level language
d. Machine language

ans: a. High level language

133. Which of the following is correct statement about pointer?
a. Pointer stores the value of variable
b. Pointer stores the address of variable
c. Pointer stores the type of variable
d. Pointer stores the size of variable

ans: b. Pointer stores the address of variable

134. Which of the following cannot be stored in a pointer?
a. int
b. float
c. double
d. string

ans: d. string

135. Which of the following header file is necessary to perform file operation in C?
a. stdio.h
b. stdlib.h
c. string.h
d. conio.h

ans: a. stdio.h

136. Which of the following header file is necessary to perform memory allocation?
a. stdio.h
b. stdlib.h
c. string.h
d. conio.h

ans: b. stdlib.h

137. Which of the following is not a keyword in C?
a. return
b. main
c. break
d. switch

ans: b. main

138. Which of the following is a bitwise operator?
a. &&
b. ||
c. <<
d. !=

ans: c. <<

139. The output of printf("%c", 65); is
a. 65
b. A
c. a
d. None of the above

ans: b. A

140. What is the output of C program?

```
void main() {
  int b = 25;
  int *p;
  p = &b;
  printf("%d %d", &b, p);
}
```

a. 25 25
b. 1234 1234
c. 25 1234
d. 1234 25

ans: c. 25 1234

141. _____ is a reserved word that has standard and predefined meanings in C.
a. Variable
b. Identifier
c. Keyword
d. None of the above

ans: c. Keyword

142. _____ is a quantity that doesn't change during program execution.
a. Variable
b. Constant
c. Data type
d. None of the above

ans: b. Constant

143. _____ is a symbolic name, which is used to store different types of data in the computer's memory and may change during the program execution.
a. Variable
b. Constant
c. Identifier
d. None of the above

ans: a. Variable

144. _____ is a symbol that instructs C to perform some operation or action on one or more operands.
a. Operand
b. Operation
c. Operator
d. None of the above

ans: c. Operator

145. _____ is something that an operator acts on.
a. Operand
b. Operation
c. Operator
d. None of the above

ans: a. Operand

146. Which of the following is used to terminate a statement in C?
a. .
b. :
c. ;
d. ,

ans: c. ;

147. Which of the following is the correct way to declare a pointer?
a. int *p;
b. int &p;
c. int p*;
d. int p&;

ans: a. int *p;

148. Which of the following is not a looping statement?
a. for
b. while
c. do-while
d. repeat

ans: d. repeat

149. Which of the following function can convert integer to string?
a. itoa
b. atoi
c. atof
d. atol

ans: a. itoa

150. Which of the following file mode is used to read and write a file?
a. r
b. w
c. r+
d. a

ans: c. r+

151. Which of the following function reads single character from file?
a. fgetc
b. fgets
c. fscanf
d. getc

ans: a. fgetc

152. Which of the following function writes a single character to a file?
a. fputc
b. fputs
c. fprintf
d. putc

ans: a. fputc

153. Which function is used to read a line from a file?
a. fgets
b. fscanf
c. fgetc
d. getc

ans: a. fgets

154. Which function is used to write a line to a file?
a. fputc
b. fputs

c. fprintf
d. putc

ans: b. fputs

155. What is the default return type of main() in C?
a. void
b. int
c. float
d. char

ans: b. int

156. Which of the following is used to ignore the rest of the loop and continue the next iteration?
a. break
b. continue
c. exit
d. return

ans: b. continue

157. Which of the following is used to terminate the loop entirely?
a. break
b. continue
c. exit
d. return

ans: a. break

158. Which of the following is the correct syntax for if-else statement?
a. if (condition) {…} else {…}
b. if condition {…} else {…}
c. if (condition) else {…}
d. if (condition) {…} else

ans: a. if (condition) {…} else {…}

159. Which of the following is correct about switch statement?
a. It can only test integer and character types
b. It can test floating point numbers
c. It can test strings
d. It can test arrays

ans: a. It can only test integer and character types

160. Which of the following is correct about break statement?
a. Break terminates the loop
b. Break skips the current iteration
c. Break returns a value
d. Break is used in functions

ans: a. Break terminates the loop

161. Which of the following is correct about continue statement?
a. Terminates the loop

b. Skips current iteration
c. Returns a value
d. Terminates program

ans: b. Skips current iteration

162. Which of the following is correct about recursion?
a. Function calling itself
b. Function calling another function
c. Function returns value
d. Function terminates program

ans: a. Function calling itself

163. Which of the following is correct about structure in C?
a. Structure is a collection of different data types
b. Structure can only contain integers
c. Structure can contain arrays only
d. Structure can only contain characters

ans: a. Structure is a collection of different data types

164. Which operator is used to access members of structure?
a) .
b) ->
c) &
d) *

ans: a) .

165. Which operator is used to access members of structure through pointer?
a) .
b) ->
c) &
d) *

ans: b) ->

166. Which of the following is correct about union?
a. Union can store only one member at a time
b. Union can store all members at a time
c. Union can store none
d. Union can store fixed values only

ans: a. Union can store only one member at a time

167. Which of the following is correct about typedef?
a. Used to define constants
b. Used to define new type name
c. Used to define functions
d. Used to define variables

ans: b. Used to define new type name

168. Which of the following is correct about enum?

a. Used to define constants
b. Used to define variables
c. Used to define pointer
d. Used to define arrays

ans: a. Used to define constants

169. Which of the following is correct about array?
a. Collection of variables of same type
b. Collection of variables of different type
c. Collection of functions
d. Collection of pointers

ans: a. Collection of variables of same type

170. Which of the following is the correct way to declare an array of 10 integers?
a. int arr[10];
b. int arr;
c. int arr();
d. int arr{};

ans: a. int arr[10];

171. Which of the following is correct about pointer arithmetic?
a. Pointers can be added or subtracted
b. Pointers cannot be added or subtracted
c. Pointers can only be multiplied
d. Pointers cannot be compared

ans: a. Pointers can be added or subtracted

172. Which of the following is correct about memory allocation functions?
a. malloc()
b. calloc()
c. realloc()
d. all of the above

ans: d. all of the above

173. Which of the following is correct about dynamic memory?
a. Memory allocated during compile time
b. Memory allocated during runtime
c. Memory is static
d. Memory cannot be allocated

ans: b. Memory allocated during runtime

174. Which of the following is correct about pointer to pointer?
a. Pointer stores address of variable
b. Pointer to pointer stores address of pointer
c. Pointer stores value of pointer

d. Pointer to pointer stores value

ans: b. Pointer to pointer stores address of pointer

175. Which of the following is correct about const pointer?
a. Pointer value can change
b. Pointer address cannot change
c. Pointer and value cannot change
d. None of the above

ans: b. Pointer address cannot change

176. Which of the following is correct about void pointer?
a. Pointer without type
b. Pointer to integer
c. Pointer to char
d. Pointer to float

ans: a. Pointer without type

177. Which of the following is correct about file pointer?
a. Pointer used to access file
b. Pointer used to access memory
c. Pointer used to access array
d. Pointer used to access string

ans: a. Pointer used to access file

178. Which of the following is correct about standard file I/O in C?
a. stdin, stdout, stderr
b. filein, fileout, fileerr
c. stdread, stdwrite, stderror
d. None of the above

ans: a. stdin, stdout, stderr

179. Which of the following is correct about fprintf()?
a. Writes formatted data to console
b. Writes formatted data to file
c. Writes unformatted data
d. Reads formatted data

ans: b. Writes formatted data to file

180. Which of the following is correct about fscanf()?
a. Reads formatted data from console
b. Reads formatted data from file
c. Reads unformatted data
d. Writes formatted data

ans: b. Reads formatted data from file

181. Which of the following is correct about feof()?
a. Returns true if end of file reached
b. Returns false if end of file reached
c. Reads character from file

d. Writes character to file

ans: a. Returns true if end of file reached

182. Which of the following is correct about fseek()?
a. Moves file pointer to desired location
b. Reads data from file
c. Writes data to file
d. Closes file

ans: a. Moves file pointer to desired location

183. Which of the following is correct about rewind()?
a. Moves file pointer to beginning
b. Moves file pointer to end
c. Moves file pointer to middle
d. Closes file

ans: a. Moves file pointer to beginning

184. Which of the following is correct about ftell()?
a. Returns current file pointer position
b. Sets current file pointer position
c. Reads data from file
d. Writes data to file

ans: a. Returns current file pointer position

185. Which of the following is correct about binary file mode?
a. Reads and writes text
b. Reads and writes binary data
c. Reads only text
d. Writes only text

ans: b. Reads and writes binary data

186. Which of the following is correct about text file mode?
a. Reads and writes binary data
b. Reads and writes text
c. Reads only binary
d. Writes only binary

ans: b. Reads and writes text

187. Which of the following is correct about fclose()?
a. Opens file
b. Closes file
c. Reads file
d. Writes file

ans: b. Closes file

188. Which of the following is correct about fflush()?
a. Clears input/output buffer

b. Reads data
c. Writes data
d. Closes file

ans: a. Clears input/output buffer

189. Which of the following is correct about temporary file?
a. File deleted automatically after program ends
b. File never deleted
c. File deleted manually
d. File stored in permanent storage

ans: a. File deleted automatically after program ends

==========

CHAPTER 4: PROGRAMMING IN C

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LONG ANSWER QUESTIONS (8 Marks)

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1. Write a C program using structure to store and display employees/students records (name, salary, post).

```
#include <stdio.h>
struct Employee {
   char name[50];
   float salary;
   char post[50];
};
int main() {
   struct Employee emp[3];
   int i;
   printf("Enter details of 3 employees:\n");
   for(i = 0; i < 3; i++) {
      printf("\nEmployee %d\n", i);
      printf("Enter Name: ");
      scanf("%s", emp[i].name);
      printf("Enter Salary: ");
      scanf("%f", &emp[i].salary);
      printf("Enter Post: ");
      scanf("%s", emp[i].post);
   }
   printf("\n--- Employee Details ---\n");
   for(i = 0; i < 3; i++) {
      printf("\nEmployee %d\nName: %s\nSalary: %.2f\nPost: %s\n", i, emp[i].name, emp[i].salary, emp[i].post);
   }
   return 0;
}
```

2) Write a C program using structure to store and display employees/students records (name,roll, address)

```
#include <stdio.h>
struct Student {
   int roll;
   char name[50];
   char address[100];
};

int main() {
   struct Student s[3];
   int i;

   printf("Enter details of 3 students:\n");

   for(i = 0; i < 3; i++) {
      printf("\nStudent %d\n", i + 1);

      printf("Enter Roll No: ");
      scanf("%d", &s[i].roll);

      printf("Enter Name: ");
      scanf("%s", s[i].name);

      printf("Enter Address: ");
      scanf("%s", s[i].address);
   }

   printf("\n--- Student Details ---\n");

   for(i = 0; i < 3; i++) {
      printf("\nStudent %d\n", i + 1);
      printf("Roll No: %d\n", s[i].roll);
      printf("Name: %s\n", s[i].name);
      printf("Address: %s\n", s[i].address);
   }

   return 0;
}
```

3. Write a C program to store data of employees/students (name, roll, marks etc.) into a file and read/display records from the file.

```
#include <stdio.h>

 struct Student {
    char name[50];
    int roll;
```

```
    float marks;
};

int main() {
    struct Student s;
    FILE *fp;
    int n, i;

    fp = fopen("student.txt", "w");

    if(fp == NULL) {
        printf("Error opening file.\n");
        return 0;
    }

    printf("Enter number of students: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("\nStudent %d\n", i + 1);

        printf("Enter Name: ");
        scanf("%s", s.name);

        printf("Enter Roll No: ");
        scanf("%d", &s.roll);

        printf("Enter Marks: ");
        scanf("%f", &s.marks);

        fprintf(fp, "%s %d %.2f\n", s.name, s.roll, s.marks);
    }

    fclose(fp);

    fp = fopen("student.txt", "r");

    if(fp == NULL) {
        printf("Error opening file.\n");
        return 0;
    }

    printf("\n--- Student Records ---\n");

    while(fscanf(fp, "%s %d %f", s.name, &s.roll, &s.marks) != EOF) {
        printf("\nName: %s\n", s.name);
        printf("Roll No: %d\n", s.roll);
        printf("Marks: %.2f\n", s.marks);
    }
```

```
    fclose(fp);

    return 0;
}
```

4. Write a C program to store data of employees/students (name, address, salary)into a file and read/display records from the file. both question are simiar and from old is gold asked 16 times

```
#include <stdio.h>
struct Employee{
  char name[50];
  char address[100];
  float salary;
};
int main(){
  struct Employee e;
  FILE *fp;
  int n,i;
  fp=fopen("employee.txt","w");
  if(fp==NULL){
    printf("Error opening file.\n");
  }
  return 0;
  printf("Enter number of employees: ");
  scanf("%d",&n);
  for(i=0;i<n;i++){
    printf("\nEmployee %d\n",i);
    printf("Enter Name: ");
    scanf("%s",e.name);
    printf("Enter Address: ");
    scanf("%s",e.address);
    printf("Enter Salary: ");
    scanf("%f",&e.salary);
    fprintf(fp,"%s %s %.2f\n",e.name,e.address,e.salary);
  }
  fclose(fp);
  fp=fopen("employee.txt","r");
  if(fp==NULL){
    printf("Error opening file.\n");
  }
   return 0;
  printf("\n--- Employee Records ---\n");
  while(fscanf(fp,"%s %s %f",e.name,e.address,&e.salary)!=EOF){
    printf("\nName: %s\nAddress: %s\nSalary: %.2f\n",e.name,e.address,e.salary);
  }
  fclose(fp);
  return 0;
}
```

5)What is function? Write a C program to find factorial of a given number using recursive. from old is gold asked 8 times

✅What is a Function?

A function is a block of code that performs a specific task. It helps in modularity, reusability, and easier debugging of programs.

```
#include <stdio.h>

int fact(int n) {
   if(n == 0 || n == 1)
      return 1;
   else
      return n * fact(n - 1);
}

int main() {
   int num;
   printf("Enter a number: ");
   scanf("%d", &num);
   printf("Factorial = %d", fact(num));
   return 0;
}
```

6) Write a C program to find factorial of a given number using user-defined function. from old is gold asked 8 times

```
#include <stdio.h>
int fact(int n) {
   int i, f = 1;
   for(i = 1; i <= n; i++) {
      f = f * i;
   }
   return f;
}

int main() {
   int num;
   printf("Enter a number: ");
   scanf("%d", &num);
   printf("Factorial = %d", fact(num));
   return 0;
}
```

7) Write a C program using function to check whether a given number is odd or even. from old is gold asked 4 times

```
#include <stdio.h>
void check(int n) {
   if(n % 2 == 0)
```

```
    printf("Even");
  else
    printf("Odd");
}
int main() {
  int num;
  printf("Enter a number: ");
  scanf("%d", &num);
  check(num);
  return 0;
}
```

8) Differentiate between structure and union with syntax / Write a program using structure and union. from old is gold asked 4 times

✅ **Difference between Structure and Union (with syntax)**

Basis	Structure	Union
Memory	Allocates **separate memory** for each member	**Shares same memory** for all members
Size	Sum of all members	Size of **largest member only**
Access	All members can be used **simultaneously**	Only **one member at a time**
Change effect	Changing one does **not affect others**	Changing one **affects all**
Keyword	struct	union

🔷 **Syntax**

Structure:

```
struct student {
  int id;
  float marks;
};
```

Union:

```
union student {
  int id;
  float marks;
};
```

Program

```
#include <stdio.h>
struct Student {
  int id;
  float marks;
};
union Data {
  int id;
  float marks;
};
int main() {
  struct Student s;
```

```
    union Data d;

    printf("Enter id and marks for structure:\n");
    scanf("%d %f", &s.id, &s.marks);
    printf("Enter id for union: ");
    scanf("%d", &d.id);
    printf("Enter marks for union: ");
    scanf("%f", &d.marks);
    printf("\n--- Structure Data ---\n");
    printf("ID: %d Marks: %.2f\n", s.id, s.marks);
    printf("\n--- Union Data ---\n");
    printf("ID: %d Marks: %.2f\n", d.id, d.marks);
    return 0;
}
```

9)What is pointer? Describe the benefits/advantages of pointer with examples. from old is gold asked 4 times

✅What is a Pointer?
A pointer is a variable that stores the memory address of another variable.
👉Example:
If `x = 10`, then a pointer `p` can store the address of `x`

🔷 **Example of Pointer shown below:**

```
#include <stdio.h>
int main() {
   int x = 10;
   int *p;
   p = &x;
   printf("Value of x = %d\n", x);
   printf("Address of x = %p\n", &x);
   printf("Value using pointer = %d\n", *p);
   return 0;
}
```

✅**Advantages / Benefits of Pointers**
1. Efficient Memory Usage
Pointers avoid copying large data.
2. Function Argument Passing (Call by Reference)
Allows modification of original value.
3. Dynamic Memory Allocation
4. Used in Arrays and Strings
Pointers help to traverse arrays efficiently.
5. Used in Data Structures
Pointers are essential in linked lists, trees, etc.

10. What are the components of a function in C? Describe call-by-value and call-by-reference / passing function parameters. from old is gold asked 3 times

Components of a Function in C

A function in C has the following main components:

1. Function Declaration (Prototype)

Declares function name, return type, and parameters.

```
int add(int,int);
```

2. Function Definition

Contains actual code (body of function).

```
int add(int a,int b){
   return a+b;
}
```

3. Function Call

Calls the function to execute.

```
sum=add(2,3);
```

Call by Value vs Call by Reference

Call by Value	Call by Reference
Copies actual value	Passes address using pointer
No change in original value	Changes original value
Uses normal variable	Uses pointer variable
Safe but less flexible	Efficient for large data

Example (Call by Value)

```
#include <stdio.h>
void change(int a){
   a=50;
}
int main(){
   int x=10;
   change(x);
   printf("%d",x);
   return 0;
}
```

Example (Call by Reference)

```
#include <stdio.h>
void change(int *a){
   *a=50;
}
int main(){
   int x=10;
   change(&x);
```

```
    printf("%d",x);
    return 0;
}
```

11) Write a C program using function to find the sum of n integer numbers. from old is gold asked 4 times

```
#include <stdio.h>

int sum(int n) {
    int i, s = 0, num;
    for(i = 0; i < n; i++) {
        printf("Enter number: ");
        scanf("%d", &num);
        s = s + num;
    }
    return s;
}

int main() {
    int n, result;
    printf("Enter how many numbers: ");
    scanf("%d", &n);
    result = sum(n);
    printf("Sum = %d", result);
    return 0;
}
```

12) Describe fscanf() and fprintf() functions. Write a program using these file handling functions. from old is gold asked 3 times

fprintf() and fscanf() Functions in C

fprintf():
- Used to write formatted data to a file
- Similar to printf(), but works with file

Syntax: fprintf(fp, "format", variables);
Example: fprintf(fp, "%d %s", id, name);

fscanf():
- Used to read formatted data from a file
- Similar to scanf(), but reads from file

Syntax: fscanf(fp, "format", &variables);
Example: fscanf(fp, "%d %s", &id, name);

C Program using fprintf() and fscanf()

```
#include <stdio.h>
```

```
int main() {
   FILE *fp;
   int id;
   char name[50];
   fp = fopen("data.txt", "w");
   if(fp == NULL) {
      printf("Error opening file.\n");
      return 0;
   }
   printf("Enter id and name: ");
   scanf("%d %s", &id, name);
   fprintf(fp, "%d %s\n", id, name);
   fclose(fp);
   fp = fopen("data.txt", "r");
   if(fp == NULL) {
      printf("Error opening file.\n");
      return 0;
   }
   fscanf(fp, "%d %s", &id, name);
   printf("ID: %d Name: %s", id, name);
   fclose(fp);
   return 0;
}
```

13) Describe file handling modes in C. Write a C program to create and write data into a file.

Asked: 3 times

File Handling Modes in C

r : Open file for reading (file must exist)
w : Open file for writing (creates new or overwrites file)
a : Open file for appending (adds data at end)
r+ : Read and write (file must exist)
w+ : Read and write (creates new or overwrites)
a+ : Read and append (creates if not exist)

Program

```
#include <stdio.h>
int main() {
   FILE *fp;
   char name[50];
   int age;
   fp = fopen("data.txt", "w");
   if(fp == NULL) {
      printf("Error opening file.\n");
      return 0;
   }
   printf("Enter name and age: ");
   scanf("%s %d", name, &age);
   fprintf(fp, "%s %d\n", name, age);
   fclose(fp);
   return 0;
}
```

14) Write a C program using structure to store student records and sort/rearrange names alphabetically. **Asked: 2 times**

```
#include <stdio.h>
#include <string.h>
struct Student {
   char name[50];
};
int main() {
   struct Student s[5], temp;
   int i, j;
   printf("Enter names of 5 students:\n");
   for(i = 0; i < 5; i++) {
      scanf("%s", s[i].name);
   }
   for(i = 0; i < 5; i++) {
      for(j = i + 1; j < 5; j++) {
         if(strcmp(s[i].name, s[j].name) > 0) {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
         }
      }
   }
   printf("Sorted names:\n");
   for(i = 0; i < 5; i++) {
      printf("%s\n", s[i].name);
   }
   return 0;
}
```

15) Write a C program using function to find the simple interest **Asked: 2 times**

```
#include <stdio.h>

float si(float p, float r, float t) {
   return (p * r * t) / 100;
}

int main() {
   float p, r, t;
   printf("Enter principal, rate and time: ");
   scanf("%f %f %f", &p, &r, &t);
   printf("Simple Interest = %.2f", si(p, r, t));
   return 0;
}
```

16) Write a C program using user-defined function to find the area of a shape (pond, football, etc.). **Asked: 2 times**

```
float area(float l, float b) {
  return l * b;
}

int main() {
  float length, breadth;
  printf("Enter length and breadth: ");
  scanf("%f %f", &length, &breadth);
  printf("Area = %.2f", area(length, breadth));
  return 0;
}
```

17) What is binary file handling in C? Explain how putw() and getw() / fprintf() and fscanf() facilitate binary input/output operations. **Asked: 2 times**

Binary File Handling in C

Binary file handling is the process of storing and reading data in binary format (machine-readable form) instead of text format. It is faster and more efficient and preserves exact data without conversion.

putw() and getw():

- Used for binary-like integer operations with files
- putw() writes an integer to file
- getw() reads an integer from file

Syntax:

```
putw(variable, fp);
variable = getw(fp);
```

Example:

```
FILE *fp;
int x = 10, y;
fp = fopen("data.dat", "wb");
putw(x, fp);
fclose(fp);
fp = fopen("data.dat", "rb");
y = getw(fp);
fclose(fp);
```

fprintf() and fscanf():

- Used for formatted text file operations (not true binary)
- fprintf() writes formatted data to file
- fscanf() reads formatted data from file

Syntax:

```
fprintf(fp, "format", variables);
fscanf(fp, "format", &variables);
```

Conclusion:

putw()/getw() are used for binary-style integer I/O, while fprintf()/fscanf() handle text-based formatted I/O.

18) Define control structure. Write a C program to find sum, difference and greater of two integers using switch case. **Asked: 1 times**

Control Structure

Control structure is a mechanism that controls the flow of execution of statements in a program. It determines the order in which instructions are executed. Examples: sequence, selection (if, switch), iteration (loop).

Program

```
#include <stdio.h>
int main() {
   int a, b, choice;
   printf("Enter two integers: ");
   scanf("%d %d", &a, &b);
   printf("1.Sum 2.Difference 3.Greater\nEnter choice: ");
   scanf("%d", &choice);
   switch(choice) {
      case 1:
         printf("Sum = %d", a + b);
         break;
      case 2:
         printf("Difference = %d", a - b);
         break;
      case 3:
         if(a > b)
            printf("Greater = %d", a);
         else
            printf("Greater = %d", b);
         break;
      default:
         printf("Invalid choice");
   }
   return 0;
}
```

19) Write a C program to delete and rename a data file using remove() and rename() commands. **Asked: 1 time**

Program

```
#include <stdio.h>

int main() {
   int r;
   r = remove("data.txt");
   if(r == 0) {
      printf("File deleted successfully\n");
   } else {
      printf("Error deleting file\n");
   }

   r = rename("old.txt", "new.txt");
```

```
    if(r == 0) {
        printf("File renamed successfully\n");
    } else {
        printf("Error renaming file\n");
    }

    return 0;
}
```

20. What is the difference between library function and user-defined function? Write a program to calculate y raised to power x using user-defined function.

Asked: 1 time

Difference between library function and user-defined function

Library Function:

- Predefined functions provided by C library
- No need to define, just use
- Example: printf(), scanf(), sqrt()

User-defined Function:

- Functions created by programmer
- Must be defined before use
- Used to perform specific tasks
- Example: sum(), fact()

Program

```
#include <stdio.h>
int power(int y, int x) {
    int i, p = 1;
    for(i = 1; i <= x; i++) {
        p = p * y;
    }
    return p;
}

int main() {
    int y, x;
    printf("Enter base and exponent: ");
    scanf("%d %d", &y, &x);
    printf("Result = %d", power(y, x));
    return 0;
}
```